Joachim Kirstein

Software Engineer

+45 51610325
jmk1589@gmail.com
jkirstein.dk
github.com/Krusovice

Summary

Software Engineer with focus on data, a background in in civil engineering and 10 years of Python experience. Independent worker and learner, taking responsibility and solving technical challenges with a strong analytical mindset. Experience in building scalable systems; from numerical modeling to full-stack web apps and ML pipelines.
With a consulting background, I take responsibility, understand business impact, and deliver high-quality solutions. I thrive in collaborative teams and I'm always eager to learn, improve and understand.

Key Experience

- **End-to-End Python Development**: 10 years of experience in Python to handle data and build object oriented apps and API's.
- **Finite Element Application in Rust**: Designed and built an FE solver from scratch, including 2D and 3D calculations with various options for boundary conditions and element selections.
- **Backend in Rust**: Webpage backend in rust, including including database handling, user authentication and serving of live metrics.
- **Machine Learning and Data Pipelines**: Built and hosted end-to-end data and ML pipelines on Azure and locally with Docker.
- **ETL and Data Architecture**: Created pipelines with Python, Rust, Apache Airflow, and Databricks. Designed storage for both structured and unstructured data.
- **Hosting & DevOps**: Developed and deployed multiple services and API's with Docker and Github Actions, hosting on Linux with a headless Raspberry Pi.
- **Frontend**: Serving frontend with React Typescript.

Selected Hard Skills

- **Processing**. Python, Rust, Databricks, Spark, SciPy, Pandas, Scikit-learn.
- **Pipelines**. Apache Airflow, (Apache Kafka).
- **Storage**. PostgreSQL, Azure Datalake Gen 2, Redis, Raspberry Pi.
- **Web & UI**. Django, Javascript, React, Rust, FastAPI.
- **Visualization**. Matplotlib, Seaborn, D3, Three.js.
- **DevOps**. Git, Docker, Github Actions.

Selected Professional Projects

The Line (Saudi Arabia), Consultant for Rambøll (2024-2025)
Lead software architect, data-, and civil engineer on analysis of 56 high-rise towers.

- Designed and implemented modular Python architecture to extract, transform, and integrate geotechnical, geometrical and structural data with commercial FE software.

- Built a custom abstraction layer on top of the FE software's REST API, dramatically improving automation, reusability, and maintainability of high-rise modeling workflows.
- Developed tools for visualization and quality assurance.

New Storstrøm Bridge, Consultant for Rambøll (2020-2021)
Lead software architect, data-, and civil engineer on analysis of 35 bridge foundations.

- Designed and implemented an optimized calculation engine for foundation settlements and stiffness response during ship collisions, executing and reporting of thousands of calculations.
- Setup pipelines on the project between the calculation software, so calculations could be automated and results stored and handled seamlessly.

Selected Personal Projects

Finite Element Calculator in Rust (2025-2026)
Developed a 2D/3D finite element solver in Rust to compute displacements and internal forces in structural models. Built modular and still adding and improving the engine. Private repo.

- Assembling of stiffness matrix, boundary conditions, force and displacement vectors, primarily using nalgebra and hashmaps for matrix operations.
- Implemented a CLI parser, enabling automation via scripting.
- Deployed in Rust via Axum and Tokio to a React based frontend.

Groundwater Prediction (2025)
A full-stack data and ML engineering project to predict groundwater levels using climate data, created and run on Azure Platform.
Repository: `https://github.com/Krusovice/dk-groundwater-predictor`.

- Designed and implemented a modular pipeline to ingest and clean climate and groundwater data from DMI's REST API.
- Applied feature engineering and trained regression models to predict groundwater levels.
- Implemented on Microsofts Azure Platform, using Databricks, Delta Lake and Factory.

Foundation Response Predictor (2025)
An ML project, to predict foundation settlements during structural loading.
Repository: `https://github.com/Krusovice/foundation-response-predictor`.

- Designed a pipeline to create, calculate and extract non-linear FE models, and store them in a database.
- Applied feature engineering and trained regression models to predict foundation response during applied forces.
- Deployed the model using FastAPI. `https://https://jkirstein.dk/foundation/` for app and documentation.

jkirstein.dk (2025-2026) Personal webpage and services hosted on Raspberry Pi. Backend in Rust with Axum+Tokio, frontend with React Typescript.
Repository: `https://github.com/Krusovice/personal-webpage`.

- Literature and media database. Few items public, full access via login.
- Stock viewer and analyser. Stocks data are scraped daily and can be fetched for comparison.
- Form for Foundation Response Predictor API, for prediction of foundation settlements.